

Supervision & Reflection Template

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What?

Reflect on what has happened in your project this month?

This month, I focused mainly on creating the appliance database that will run the energy estimation features of my project. I completed the design of a unified JSON schema for all appliance types and researched accurate wattage values across most of the commonly found household appliances. I created the full JSON dataset, which is now ready for backend usage. I also finished an architectural decision for Sprint 3: instead of splitting appliances into two separate files, I moved forward with a single JSON file using a `usageType` field to distinguish continuous devices from per-use devices. This work became a new task added to my sprint plan and was fully completed this month.

So What?

Consider what that meant for your project progress. What were your successes? What challenges remain?

Getting the authentication system working was a major step because the entire private side of the app depends on it. Until now, the lack of a reliable login system was blocking features like Dashboard, Map My House, and My Appliances. Implementing JWT authentication, password hashing and secure routes means the foundation of the whole platform is now stable. It also forced me to work through real challenges, everything from Java/Maven version issues to circular dependencies inside Spring Security. Everything was a leaning curve and plenty of research was needed.

Finishing the appliance database also matters because it ties into the energy estimation logic described in the system overview. With the JSON dataset complete, I can now move on to the backend services that will calculate energy usage. October was pretty successful for my final year project.

Now What?

What can you do to address outstanding challenges?

Next month, I will start integrating the appliance database into the backend by building the models, services, and endpoints needed to retrieve appliance details and perform usage calculations. I will also use my frontend diagrams and frameworks and transfer them to my final version. Also hoping that it is seamless to make the backend work with it.

Student Signature

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